PROJECT REPORT

Design and Analysis of a Hybrid Parallel Prefix Adder Using Han-Carlson and Ladner-Fischer Architectures

Low-power 4-bit adder at 32 nm in six logic styles, plus a 2-bit reversible quantum adder

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Key Results

60.17%	**24.00%**	**35.00%**	**3 + 3**
lower power than CMOS (0.235 mW vs 0.590 mW)	lower delay than CMOS (38 ns vs 50 ns)	smaller area than CMOS (6.5 vs 10.0 µm²)	Toffoli + CNOT gates in the 2-bit adder

Comparison of the proposed FSGDI hybrid adder (4-bit, 32 nm, Microwind) with the CMOS version of the same adder. The reversible adder is a 2-bit functional proof of concept; no power figures are claimed for it.

Project at a Glance

Domain	VLSI design, low-power digital arithmetic, reversible/quantum logic
Tools	DSCH 3.5, Microwind 3.1, IBM Quantum Composer, Qiskit (OpenQASM 2.0)
Skills shown	Transistor-level design, 32 nm layout and DRC, post-layout power/delay/area analysis, Boolean-to-quantum gate mapping
Technology	32 nm CMOS, GDI-family logic styles, 2-bit reversible circuit

Summary

High-speed arithmetic circuits are essential in modern VLSI systems, where adder performance strongly influences overall computational efficiency. This project presents a hybrid Parallel Prefix Adder (PPA) that combines the Han-Carlson and Ladner-Fischer prefix structures. Han-Carlson keeps wiring and fan-out low, while Ladner-Fischer gives low logic depth and fast carry propagation.

The 4-bit design was built at transistor level in six logic styles (CMOS, PTL, DPTL, GDI, MGDI and FSGDI) using DSCH 3.5, with 32 nm layouts and power, delay and area analysis in Microwind 3.1. The same Boolean equations were then mapped to reversible quantum gates (CNOT and Toffoli), and a 2-bit version was simulated in IBM Quantum Composer and Qiskit.

1. Introduction and Motivation

Binary adders sit at the heart of arithmetic logic units, digital signal processors, microprocessors, cryptographic engines and AI accelerators. Because so many operations reduce to addition, adder speed and energy strongly affect overall system performance.

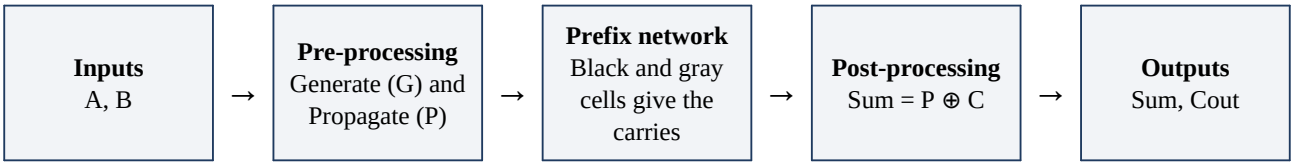
In a ripple-carry adder every bit waits for the carry from the bit before it, so delay grows linearly with word length. Parallel prefix adders remove this bottleneck by computing all carries in parallel through a tree of prefix operations, reaching logarithmic depth. As technology scales, leakage, wiring complexity, switching power and fan-out also matter, so both the prefix structure and the transistor-level logic style are important.

Project Objectives

- Design a 4-bit hybrid Han-Carlson + Ladner-Fischer PPA that balances speed, area and routing.
- Compare six logic styles (CMOS, PTL, DPTL, GDI, MGDI, FSGDI) at 32 nm for power, delay and area.
- Map the same equations to CNOT and Toffoli gates and verify a 2-bit reversible adder in IBM Quantum Composer and Qiskit.

2. Background: How a Parallel Prefix Adder Works

Every PPA follows the same three stages; only the prefix network differs between architectures.



- **Pre-processing:** Gi = Ai·Bi (the bit creates a carry) and Pi = Ai ⊕ Bi (the bit passes a carry on).
- **Prefix network:** neighbouring pairs merge as (Gi, Pi) (Gj, Pj) = (Gi + Pi·Gj, Pi·Pj). A black cell gives both outputs; a gray cell gives only the group generate.
- **Post-processing:** the sum bit is Si = Pi ⊕ Ci, where Ci is the carry into bit i.

3. Han-Carlson vs Ladner-Fischer and the Research Gap

Property	Han-Carlson	Ladner-Fischer
Prefix structure	Sparse: carries for alternate bits first, then filled in	Balanced tree with the fewest levels
Logic depth	Slightly higher (one extra level)	Lowest (log2 n levels)
Wiring and fan-out	Low congestion, balanced fan-out	High fan-out on later nodes
Weakness	Extra level adds delay	Heavy loading slows nodes

Neither structure alone gives the best trade-off between speed, area, routing and power. Earlier DSCH/Microwind studies focused on single prefix structures, full adders and ALUs in individual logic styles. A hybrid prefix adder compared across all six styles at one node, and realised in reversible quantum gates, was left as future scope in earlier work. This project addresses that gap.

4. Proposed Hybrid Architecture (4-bit)

The first carry stages follow the Han-Carlson pattern, which limits wiring and fan-out. The later stages follow the Ladner-Fischer tree, which distributes carries in a balanced way. Together they shorten the critical carry path while keeping hardware overhead low.

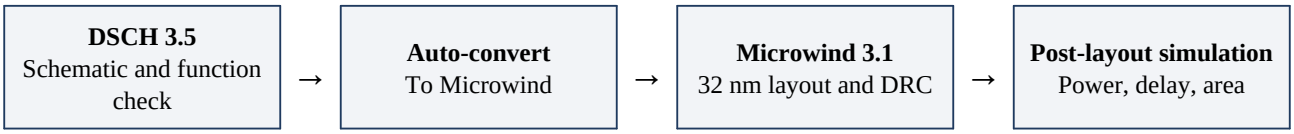
Stage	Boolean equations
1. Pre-processing	Gi = Ai·Bi, Pi = Ai ⊕ Bi (i = 0 to 3)
2. Han-Carlson stage	G1:0 = G1 + P1·G0, P1:0 = P1·P0 G3:2 = G3 + P3·G2, P3:2 = P3·P2
3. Ladner-Fischer stage	C1 = G0, C2 = G1:0, C3 = G2 + P2·G1:0, C4 = G3:2 + P3:2·G1:0
4. Post-processing	Si = Pi ⊕ Ci (C0 = 0, so S0 = P0), Cout = C4

5. Six Logic Styles Compared

The same architecture was implemented in every style under identical technology (32 nm), tools and operating conditions, so the comparison is fair.

Style	Idea	Strength	Limitation
CMOS	Complementary pull-up and pull-down	Full swing, high noise immunity	Most transistors: larger area and power
PTL	Signals pass through switch transistors	Fewer transistors, compact	Threshold drop, degraded swing
DPTL	Complementary NMOS and PMOS pass paths	Better voltage restoration	More routing overhead
GDI	Many functions from one compact cell	Low transistor count and power	Reduced output swing
MGDI	GDI with modified connections	Better swing, still compact	Mostly shown on small blocks
FSGDI	GDI with full-swing restoration	Compact and full swing	Newest; limited prior use on PPAs

6. Design Flow and Tools



Each design was verified with input vectors and timing waveforms in DSCH, converted to a Microwind-compatible file, laid out at 32 nm, checked with design-rule checking (DRC) and re-simulated after layout. Power, delay and area were read from Microwind under the same conditions for every style.

7. Results: 4-bit Hybrid Adder (Conventional Implementation)

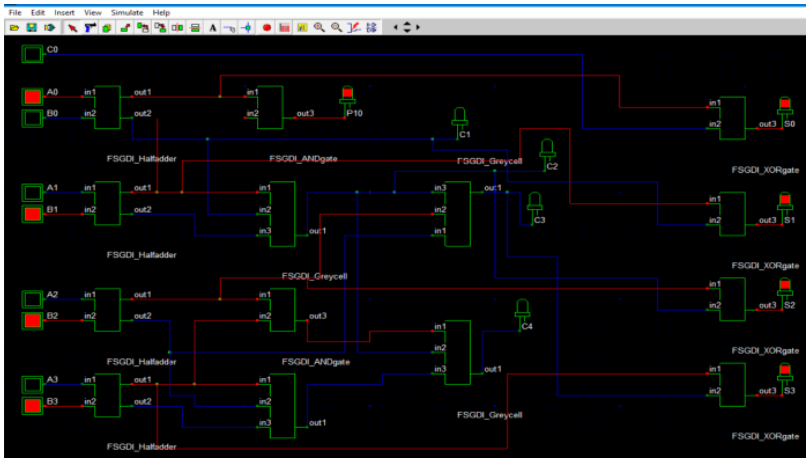


Fig. 1 FSGDI hybrid adder schematic (DSCH)

Schematic (DSCH). FSGDI half-adder cells form the pre-processing stage, FSGDI AND and gray cells form the prefix stages, and FSGDI XOR gates give sum bits S0 to S3 with carries C1 to C4. Simulation confirmed correct propagate, generate, carry and sum signals with full-swing outputs.

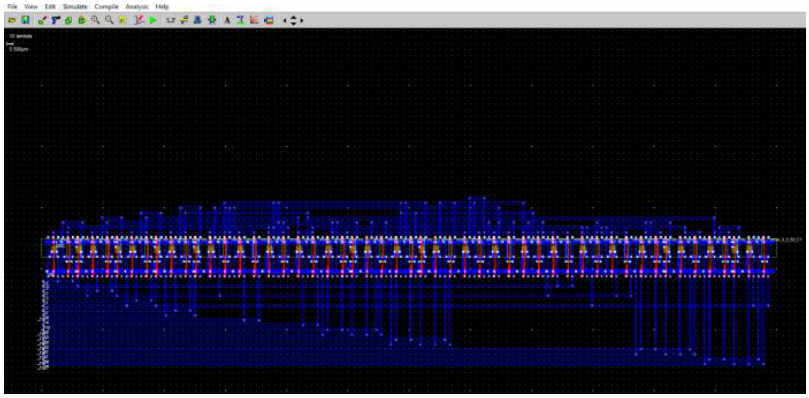


Fig. 2 Physical layout of the FSGDI hybrid adder

Layout (Microwind, 32 nm). The DSCH design was converted automatically, so connectivity is preserved with no manual redesign. Fewer transistors give the most compact layout of the six styles (about 6.5 µm²). DRC was run before post-layout simulation.

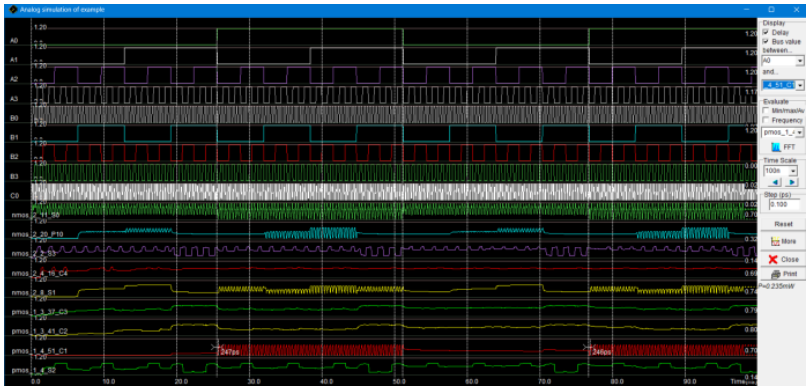


Fig. 3 Power waveforms of the FSGDI hybrid adder

Power analysis (Microwind). All input combinations were applied and average power reported. The FSGDI hybrid adder dissipates the least of the six styles: 0.235 mW. A lower transistor count means lower switching capacitance and therefore lower dynamic power.

8. Comparative Performance Analysis

Table 1 and Fig. 4 summarise all six styles under identical 32 nm conditions. FSGDI is best on every metric while keeping full output swing; MGDI is second. Delay and area values are approximate Microwind estimates.

Logic style	Power (mW)	Delay (ns)	Area (µm²)	Output swing
CMOS	0.590	50.00	10.00	Full
PTL	0.389	45.00	8.00	Degraded
DPTL	0.414	46.00	8.50	Improved
GDI	0.387	44.00	7.80	Partial
MGDI	0.340	42.00	7.50	Improved
FSGDI (proposed)	**0.235**	**38.00**	**6.50**	**Full**

Table 1 Power, delay, area and output swing of the hybrid adder in six logic styles (32 nm)

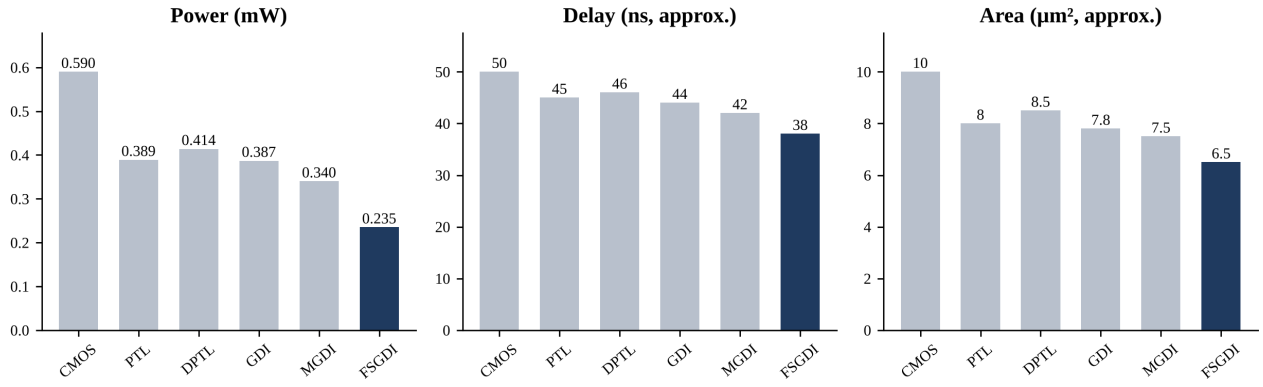


Fig. 4 Power, delay and area by logic style; FSGDI (dark blue) is lowest in all three

Compared with	Power reduction	Delay reduction	Area reduction
CMOS	60.17%	24.00%	35.00%
PTL	39.59%	15.56%	18.75%
DPTL	43.24%	17.39%	23.53%
GDI	39.28%	13.64%	16.67%
MGDI	30.88%	9.52%	13.33%

Table 2 Improvement of FSGDI over each style = (existing − proposed) / existing × 100

Reading the results. Even against the optimised MGDI version, FSGDI saves a further 30.88% power, 9.52% delay and 13.33% area. The gains come from the lower transistor count, which cuts switching capacitance and routing, and from full-swing restoration, which keeps signals reliable through the cascaded prefix stages.

9. Reversible Quantum Implementation (2-bit)

Quantum circuits must be reversible, so AND maps to the Toffoli gate (CCX) and XOR to the CNOT gate (CX). The circuit uses six qubits: a1a0, b1b0 and two ancillas g0, g1. The OR in G1:0 = G1 + P1·G0 becomes an XOR because G1 and P1·G0 are never both 1, which keeps the adder at 3 Toffoli and 3 CNOT gates.

Stage	Gate	Result
Generate	Toffoli (a0, b0 → g0)	G0 = A0·B0
Generate	Toffoli (a1, b1 → g1)	G1 = A1·B1
Propagate	CNOT (a0 → b0)	b0 = P0 = S0
Propagate	CNOT (a1 → b1)	b1 = P1
Carry	Toffoli (b1, g0 → g1)	g1 = G1 ⊕ P1·G0 = Cout
Sum	CNOT (g0 → b1)	b1 = P1 ⊕ G0 = S1

Table 3 Gate sequence of the 2-bit reversible adder

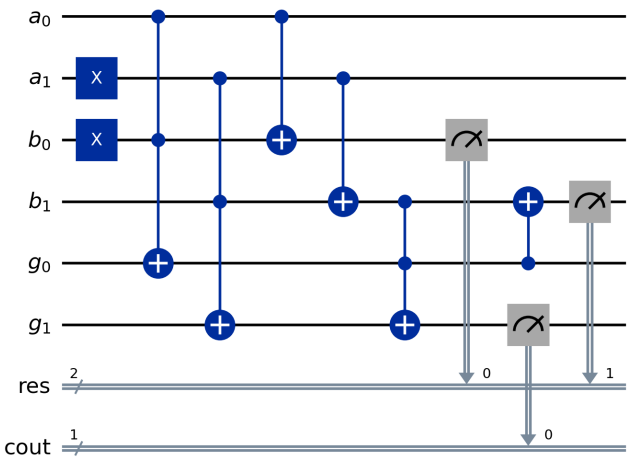


Fig. 5 2-bit reversible adder built in IBM Quantum Composer (OpenQASM 2.0) and drawn with Qiskit

Verification: all 16 inputs, output shown as (Cout Sum), checked in Qiskit Aer. For example, A = 10 and B = 01 gives Cout = 0, Sum = 11 (2 + 1 = 3) with 100% probability.

A \ B	00	01	10	11
00	0 00	0 01	0 10	0 11
01	0 01	0 10	0 11	1 00
10	0 10	0 11	1 00	1 01
11	0 11	1 00	1 01	1 10

Table 4 Truth table of the 2-bit reversible adder

Note on scope. At 2 bits there is only one prefix node (G1:0), so Han-Carlson and Ladner-Fischer coincide. This circuit is a verified proof of concept of the 4-bit equations, not yet a full hybrid structure. The 4-bit reversible version, where the two structures differ, is the natural next step.

10. Conclusion and Future Work

- A hybrid Han-Carlson + Ladner-Fischer PPA was designed and compared in six logic styles at 32 nm. FSGDI achieved 0.235 mW, 38 ns and 6.50 μm²: 60.17% less power, 24.00% less delay and 35.00% less area than CMOS, with full output swing.
- The same equations were mapped to 3 Toffoli and 3 CNOT gates, and a 2-bit reversible adder was verified on all 16 inputs.
- Next steps: build and simulate the 4-bit reversible adder; extend to wider words and advanced nodes (22 nm, FinFET); test under process, voltage and temperature variation; integrate into ALU and MAC units.

References

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[4] Qiskit contributors, "Qiskit: An Open-source Framework for Quantum Computing," 2023.